

BILAL AHMED

DevOps Engineer | Cloud & DevSecOps | Kubernetes | CI/CD

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PROFESSIONAL SUMMARY

Software Engineering graduate with hands-on experience building and deploying **cloud-native applications and Kubernetes environments** using **AWS, Linux, Docker, GitHub Actions, Terraform, Kubernetes, Helm, and k3s**. Experienced in implementing **CI/CD pipelines, container security, infrastructure as code, Kubernetes administration, monitoring, dashboards, and alerting**. Strong troubleshooting experience across Linux, networking, containers, cloud infrastructure, and application deployments. Seeking a **DevOps Intern or Junior DevOps Engineer** role.

TECHNICAL SKILLS

- **Cloud & Infrastructure:** AWS EC2, S3, IAM, Terraform, Infrastructure as Code
- **Containers & Orchestration:** Docker, Kubernetes, Kind, k3s, Helm, Traefik
- **CI/CD & Automation:** GitHub Actions, Docker Hub, Bash, Python, SSH, GitHub Secrets
- **Monitoring & Observability:** Prometheus, Grafana, Kubernetes Metrics, Node Exporter, Alerting, PromQL
- **Linux & Networking:** Ubuntu/Linux, Bash, SSH, File Permissions, Processes, TCP/IP, HTTP/HTTPS, DNS
- **Security:** DevSecOps Fundamentals, Non-root Containers, Linux Capabilities, Privilege Escalation Controls, Kubernetes RBAC, NetworkPolicy
- **Development:** Python, FastAPI, REST APIs, JavaScript, TypeScript, React.js
- **Databases & Tools:** MongoDB Atlas, MySQL, SQLite, Postman, MobaXterm

PROJECTS

DevOps Cloud Infrastructure & CI/CD — Project 1 | [GitHub](#)

- Built and deployed a **containerized web application on AWS EC2** using Docker and Nginx.
- Implemented **GitHub Actions CI/CD** for application validation and Docker image build/push workflows.
- Managed container images using **Docker Hub** and secured CI/CD credentials with GitHub Secrets.
- Provisioned AWS infrastructure using **Terraform**, including an S3 bucket, and practiced Infrastructure as Code workflows.
- Configured and troubleshoot Linux, SSH, Docker, AWS, Git, and cloud deployment environments.

Technologies: AWS EC2, S3, IAM, Terraform, Docker, Docker Hub, GitHub Actions, Linux, Bash

Kubernetes Deployment & Administration — Project 2 | [GitHub](#)

- Built a **3-node Kubernetes cluster using Kind** and deployed a containerized web application.

- Configured Deployment, Service, Ingress, ConfigMap, Secret, PVC, ServiceAccount, RBAC, and NetworkPolicy.
- Implemented **Horizontal Pod Autoscaling (HPA)** with 3–6 replicas based on CPU utilization.
- Configured readiness/liveness probes and CPU/memory requests and limits.
- Validated workload health, networking, scaling, and Kubernetes resources using kubectl.

Technologies: Kubernetes, Kind, Docker, kubectl, YAML, Ingress, HPA, RBAC, NetworkPolicy

CI/CD Kubernetes Deployment on AWS — Project 3 | [GitHub](#)

- Built a **production-style CI/CD pipeline** using GitHub Actions for automated testing, Docker image builds, and Docker Hub publishing.
- Published versioned Docker images using **latest and Git SHA tags**.
- Deployed validated Git commits to **k3s on AWS EC2 using Helm** and automated SSH-based continuous delivery.
- Configured **Traefik Ingress, HPA, health probes, resource limits, and security contexts**.
- Implemented container hardening using non-root execution, dropped Linux capabilities, and disabled privilege escalation.

Technologies: Node.js, Express, Docker, GitHub Actions, Docker Hub, AWS EC2, k3s, Kubernetes, Helm, Traefik

Kubernetes Observability & Monitoring — Project 4 | [GitHub](#)

- Deployed a complete Kubernetes observability stack using **Prometheus and Grafana through Helm**.
- Built a **13-panel Grafana dashboard** monitoring CPU, memory, disk, network, pod health, deployments, and container resource usage.
- Implemented **PromQL-based monitoring** for Kubernetes workloads and infrastructure metrics.
- Configured and tested **5 Grafana alert rules** covering CPU, storage, pod restarts, deployment availability, and Prometheus target health.
- Integrated **Gmail SMTP alert notifications** and validated firing/resolved alert workflows using controlled failure scenarios.

Technologies: Prometheus, Grafana, Helm, Kubernetes, k3s, PromQL, Node Exporter, kube-state-metrics, Alerting

EDUCATION

Bachelor of Science in Software Engineering

University of Mianwali — Mianwali | **2022–2026**

CGPA: **2.83 / 4.00**

HSSC — Pre-Engineering

The Superior Group of Colleges — Quaidabad | **2020–2022**

Marks: **815/1100**